EARTHPRESS · TRACEABILITY MATRIX

EarthPress traceability matrix

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/earthpress		
Source	docs/03b-traceability.md @ 748a1fc		

REVISION HISTORY

Version	Date	Author	Change
0.4	2026-10-29	Amish Chadha	Generated by .kit/trace.py
0.3	2026-10-29	Amish Chadha	Generated by .kit/trace.py
0.2	2026-10-29	Amish Chadha	Generated by .kit/trace.py
0.1	2026-10-25	Amish Chadha	Generated by .kit/trace.py

EarthPress traceability matrix

Generated by `python .kit/trace.py` from the user requirements, product requirements, calculations, build plan and BOM. Do not edit by hand. 16 user requirements, 16 product requirements, 0 gaps.

Table 1. User requirement to product requirement to evidence to verification.

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-001	PR-001	Block dimensions	EPR-CAL-001, EPR-BLD-001	Analysis (CAL-001 sections 5 and 6), TRL 3; gauge checks on a production run, TRL 4	Met on paper: 90.0 mm at the crank stop with the fill within 0 to +2.7 %; mold bulge 0.08 mm
UR-002	PR-002	Compaction pressure	EPR-CAL-001, EPR-BLD-001	Analysis (CAL-001 sections 3 to 5), TRL 3; load cell, TRL 4	At risk**: 2.00 MPa with two operators and a weighed fill; one operator at 500 N stalls at 0.53 MPa; rests on the assumed soil law (STM S6)
UR-002, UR-016	PR-003	Block strength with suitable soil	EPR-CAL-001, EPR-STM-001	Literature, TRL 3; block tests, TRL 4	To verify at TRL 4
UR-003	PR-004	Output	EPR-CAL-001, EPR-BLD-001	Analysis (CAL-001 section 9), TRL 3; timed trials, TRL 4	At risk**: 296 a day on an assumed 85 s cycle (shortfall accepted by Amish, 2026-09-25)
UR-004	PR-005	Operator force and grip height	EPR-CAL-001, EPR-BLD-001, EPR-STM-001	Analysis (CAL-001 sections 3, 4 and 4b), TRL 3; force at the grip, TRL 4	Met on paper with two operators: 334 N each (base soil), 405 N each (softer design soil); grip 0.86 to 1.89 m; peak at 1.16 m; eject 469 N for one
UR-005	PR-006	Garage-buildable	EPR-CAL-001	Inspection of the model, drawings and build plan (BLD-001), TRL 3	Met by design
UR-006	PR-007	Movable	EPR-CAL-001, EPR-BLD-001, EPR-STM-001	Analysis (CAL-001 section 8, kit 1.9.1 model), TRL 3; weighing, TRL 4	Not met**: 203.8 kg with every bush, nut, washer and weld bead (3.8 kg over; decision D2, proposed, awaiting Amish); heaviest piece 44.5 kg (met); plan 1.28 x 0.58 m (met)
UR-007, UR-015	PR-008	Durability	EPR-CAL-001	Analysis (CAL-001 section 6, fatigue), TRL 3; endurance log, TRL 4	To verify at TRL 4: welded range 89 MPa against 243 MPa, pin range 234 MPa against about 350 MPa; wear needs soil data
UR-008	PR-009	Soil test kit	EPR-CAL-001, EPR-STM-001	Inspection of the written procedure (BLD-001), TRL 3; comparison with laboratory grading, TRL 4	Met by design; accuracy to verify at TRL 4
UR-010	PR-010	Low stabilizer use	EPR-CAL-001	Literature, TRL 3; block tests, TRL 4	To verify at TRL 4 (design at 5 %, 0.33 kg a block)

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-011	PR-011	Embodied carbon	EPR-CAL-001, EPR-STM-001	Analysis (CAL-001 section 10, cited factors), TRL 3	Met on paper: 7.6 % (0.35 t against 4.55 t for a small house)
UR-012	PR-012	Safe operation	EPR-CAL-001, EPR-BLD-001, BOM	Analysis and design review (CAL-001 section 6), TRL 3; proof-load test, TRL 4	Met on paper: rest catch and end pawl; lid and latch rated 173 kN; linkage guard
UR-009	PR-013	Affordable	EPR-CAL-001	Analysis of the priced BOM (`bom/bom.csv`), TRL 3	Over target**: USD 542 (indicative), USD 42 over; savings in EPR-DEC-001
UR-013	PR-014	Open and documented	EPR-CAL-001	Review by an outside fabricator, TRL 4	To verify at TRL 4
UR-014	PR-015	Human powered	EPR-CAL-001, EPR-STM-001	Inspection of the model and BOM, TRL 3	Met by design
UR-012, UR-007	PR-016	Structure at the design load	EPR-CAL-001	Analysis (CAL-001 sections 6 and 6b), TRL 3; proof-load test, TRL 4	Met on paper after fix F1 (EPR-DEC-001): lowest factor 1.29; the 35 mm C45 pins did not meet it

Gaps

None.